



Acid gas-induced fabrication of hydrophilic carbon nitride with dual defects for boosting photocatalytic H₂O₂ generation and contaminants degradation

Na Li, Chen Wang, Wei Gao, Wenli Li, Binrong Li^{*}, Xuedong Wang^{*}

National and Local Joint Engineering Laboratory of Municipal Sewage Resource Utilization Technology, School of Environmental Science and Engineering, Suzhou University of Science and Technology, Suzhou 215009, PR China

ARTICLE INFO

Editor: Xiyang Xu

Keywords:

Defected carbon nitride
Hydrogen peroxide synthesis
Photocatalytic oxygen reduction
Superoxide radical
Degradation

ABSTRACT

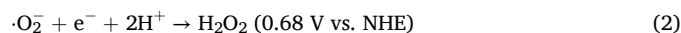
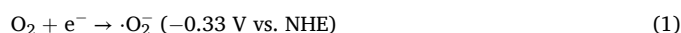
An acid gas-induced strategy was proposed to fabricate a series of PCNs with dual-defects and surface hydroxyl group (DDH-PCNs) for photocatalytic H₂O₂ generation and trace organic contaminants (TrOCs) degradation. The Nv can accelerate the O₂ adsorption and activation by formation a bridge C–O–O–C model. Accompanied cyano groups provide the asymmetric electron distribution for carriers' separation. Moreover, the surficial hydroxyl groups provide more protons for photocatalytic H₂O₂ production. As a result, the DDH-PCNs showed an excellent photocatalytic H₂O₂ rate of 5554.6 μmol g^{−1} h^{−1} under full spectrum irradiation, and it can also degrade SMX (100 μg/L) completely in 5 min. Notably, the optimal DDH-PCN(1.6) can simultaneously achieve photocatalytic H₂O₂ production and SMX degradation in a wastewater/alcohol system. The rapid formation of the intermediate superoxide radical (2.44 min^{−1}) is the key to above dual functions. Our findings provide profound insights into ·O₂[−] manipulation and development of dual functional photocatalytic system towards environmental application.

1. Introduction

With the acceleration of industrialization, the increase in energy demand and the deterioration of the global environment have become two major challenges faced by mankind in recent decades. Green H₂O₂, as a new energy carrier in fuel cells, has been widely concerned in energy applications because of its high energy density [1–3]. Specially, it is emerging in the field of wastewater treatment due to its non-toxic by-product and strong oxidation [4,5]. However, conventional anthraquinone and electrocatalytic oxygen reduction methods consume large amounts of energy and generates waste for H₂O₂ production, and are highly unsafe during the transportation and storage of H₂O₂. In addition, antibiotic micropollutants have been widely detected in aquatic environment in quantities ranging from ppb to ppm levels recently, they are thus often referred to as trace organic contaminants (TrOCs) [6,7]. Even extremely low concentration, SMX can be potentially harmful to the aquatic ecosystem and human health by biomagnification, and is a carcinogen. Nevertheless, traditional wastewater treatment methods are ineffective in completely degrading these TrOCs [8].

In the context of above two themes, H₂O₂ photosynthesis and TrOCs degradation is the most sustainable and promising strategy, making a significant contribution to a greener and cleaner environment.

Photocatalysis is such a two-in-one strategy for both H₂O₂ production and pollutants degradation by utilizing solar energy and molecular oxygen, which is completely depends on reactive oxygen species (ROS), such as singlet oxygen (¹O₂), superoxide radical (·O₂[−]), and hydroxyl radical (·OH) [9]. Notably, ·O₂[−] acts as a negatively charged radical, which not only further obtains electrons to generate H₂O₂, but also directly degrades pollutants (Eqs. (1)–(3)) [10]. Therefore, manipulating the yield of ·O₂[−] radicals is key toward the deployment of this dual functional photocatalysis strategy.



Polymeric carbon nitride (PCN) is an attractive visible-light responsive photocatalyst, showing non-metallic properties, stability, and suitable band gap structures, which has been widely investigated in the activation of molecular oxygen. However, limited by severe recombination of photogenerated carriers and a lack of effective active sites, utilization of O₂ and photogenerated electrons is inefficient. Fortunately, defect engineering has high efficiency in modulating charge carrier transfer, and establishing surface functional group of

^{*} Corresponding authors.

E-mail addresses: libr@usts.edu.cn (B. Li), zjuwxid@163.com (X. Wang).

<https://doi.org/10.1016/j.seppur.2024.129978>

Received 4 August 2024; Received in revised form 30 September 2024; Accepted 2 October 2024

Available online 3 October 2024

1383-5866/© 2024 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

photocatalysts. Appropriate defect structures, such as vacancies (C [11] or N vacancies [12,13]) and surface functional groups ($-\text{C}\equiv\text{N}$ [14–16], $-\text{OH}$ [17], or $-\text{NH}_2$ [18]) can synergistically improve photocatalytic redox reactions. Among them, the $-\text{C}\equiv\text{N}$ groups with strong electron-withdrawing characteristics can lead to the delocalization and concentration of isolated valence electrons in π -conjugated heterocycles [19]. Feng et al. have pointed out that the alkali metal reagents or borohydrides assistant polycondensation process is able to insert N vacancies, narrowing the bandgap and adsorb and activate O_2 for enhancing ORR [20]. On the other hand, TrOCs with extremely low concentration are difficult to effectively contact with bulk photocatalysts in two-phase reaction systems, which results in inadequate degradation of TrOCs. Regulating the surface functional groups of catalysts to achieve good dispersibility may be a promising avenue [17]. The surface hydroxyl components can regulate the wettability of the catalyst, thereby enhancing its dispersibility and accelerating mass transfer between solid and liquid. Until now, it is challenging to construct PCN multi-defects structure by a facile strategy. More importantly, there is still a lack of in-depth research on the effect of various configuration in photocatalysis.

In this work, we constructed a dual-defect PCN with hydroxyl groups (DDH-PCNs) by using a one-step acid gas tailoring strategy, and performed systematical research on reaction mechanism through experiments and theoretical calculations. Dual defects ($-\text{C}\equiv\text{N}$ groups and N vacancies) play a crucial role in O_2 adsorption/activation (formation of $\text{C}-\text{O}-\text{O}-\text{C}$ bridge model) and charge separation. Furthermore, the acidic SO_2 gas endows PCN abundant surface $-\text{OH}$ groups, which directly participated in ORR and promoted the contact with trace SMX. Consequently, DDH-PCNs not only selectively produce H_2O_2 with $5554.6 \mu\text{mol g}^{-1} \text{h}^{-1}$ under full spectrum, but also can completely degrade trace pollutants less than 1.0 mg/L . More importantly, the optimal DDH-PCN(1.6) can simultaneously achieve photocatalytic H_2O_2 production and SMX degradation in a wastewater/alcohol system, which is promising for the treatment and recycling of alcohol-containing organic wastewater. This study provides a new insight in the fields of energy and environmental remediation.

2. Experimental section

2.1. Preparation of photocatalysts

The specific production steps for DDH-PCNs are as follows: 6.0 g of melamine and $X \text{ g}$ of sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$, $X = 0.4, 0.8, 1.2, 1.6$ and 2.0) were fully grinded to a uniformly powder. The powder was transferred into a porcelain crucible and calcined at 500°C for 4 h in air atmosphere with heating rate of $5^\circ\text{C}/\text{min}$. The products, denoted as DDH-PCN(X), were cleaned by deionized water and dried for further using. As a comparison, PCN is calcined by using melamine under the same conditions. The PCN- $\text{K}_2\text{S}_2\text{O}_3$, PCN- Na_2SO_4 , PCN- $(\text{NH}_4)_2\text{S}_2\text{O}_3$, PCN- $(\text{NH}_4)_2\text{SO}_4$ and PCN- NaCl prepared by replacing $\text{Na}_2\text{S}_2\text{O}_3$ with corresponding salts without other change.

2.2. Photocatalytic H_2O_2 production

Firstly, 50 mg of as-prepared samples were dispersed in 50 mL of ultrapure water containing 5% of anhydrous ethanol. Above suspension was stirred under continuous air injection. A 300 W xenon lamp ($\lambda > 420 \text{ nm}$) was used as the visible light source. The reaction temperature was controlled at 25°C by condensing unit. After the photocatalytic reaction, the iodometry is used to determine the H_2O_2 concentration in filtered solution. Specifically, the mixture solution of 1.0 M HAc and 1.0 M NaHAc was used as solution A, 1.0 M of KI and 0.01 M of ammonium molybdate was solution B and solution C, respectively. $100 \mu\text{L}$ of solutions A, B, and C were added into filtered sample. The absorbance at 350 nm was obtained by using UV-vis spectrophotometry, and the linear relationship between H_2O_2 concentration and UV-vis absorption

intensity is shown in Fig.S1.

2.3. Photocatalytic degradation experiment

The experiment of photocatalytic degradation of SMX requires 50 mg of as-prepared samples were dispersed in 50 mL of SMX solution ($100 \mu\text{g/L}$, 1 mg/L , 5 mg/L , or 10 mg/L). After the light is turned on, samples are taken at various intervals. For the detection of pollution during the degradation process, a high-performance liquid phase is used, with a mobile phase of 25% acetonitrile and 75% formic acid water (0.1%), and a flow rate of $1 \text{ mL}/\text{min}$. We did the experiments to investigate the effect of higher or lower dosage of DDH-PCNs on degradation of SMX (Fig.S2).

3. Results and discussion

3.1. Preparation and microstructure of the photocatalysts

The synthetic process of DDH-PCN(1.6) prepared by a one-step calcination method is shown in Fig. 1a. Under high temperature, the thiosulfate was decomposed to produce acidic SO_2 gas ($2\text{Na}_2\text{S}_2\text{O}_3 + 3\text{O}_2 = 2\text{Na}_2\text{SO}_4 + 2\text{SO}_2$), which alters the thermal polycondensation process of melamine, resulting in the change of structures. As shown in Fig. S3–S4 and Fig. 1c–d, the DDH-PCN(1.6) shows a smaller and thinner structure than that of bulk PCN, which is probably due to the “cutting effect” of abundant acidic SO_2 gas. Moreover, it is well known that the NH_3 will be produced through deamination in thermal polymerization of melamine to form PCN. The acid SO_2 gas will react with NH_3 ($2\text{NH}_3 + \text{SO}_2 + \text{H}_2\text{O} = (\text{NH}_4)_2\text{SO}_3$ or $\text{NH}_3 + \text{SO}_2 + \text{H}_2\text{O} = \text{NH}_4\text{HSO}_3$), thus promoting the deamination and probably resulting in the formation of N vacancies, which can be proved by following analysis. Element mapping (Fig. 1b and Fig.S5) shows almost no element S, which strongly proves that SO_2 gas has escaped after modification. The detection of Na element in DDH-PCN(1.6) suggests the introduction of Na ions into the PCN framework. Interestingly, the DDH-PCN(1.6) still maintains a turbid state after 15 days and no obvious deposition is observed (Fig.S6), which is attributed to its stabilization by electron interaction. The zeta potential results also demonstrated that DDH-PCN(1.6) has superior dispersity in water compared to bulk PCN (Fig.S7). Above findings indicate that the acid gas tailoring strategy can alter the microscopic morphology and dispersibility, which may be beneficial for its photocatalytic performance.

3.2. Characterization of defective structure

The crystal structure of synthesized PCNs is compared through X-ray diffraction (XRD) spectra. As shown in Fig. 2a, the sharp characteristic peak (002) near 27.6° is assigned to the interlayer stacking between conjugated aromatic systems, while another negligible characteristic peak (100) at 13.1° is portioned to the ordered structure of the tri-s-triazine units in PCN [19,21]. The (002) peak significantly weakens and widens with the addition of $\text{Na}_2\text{S}_2\text{O}_3$, which clearly manifests the disruption and exfoliation of DDH-PCN(1.6). Almost disappeared (100) peak implies the change of ordered structure and formation of defects [22]. In addition, two small peaks appear at 7.5° and 10.4° in DDH-PCN(2.0), which are due to the poly (heptazine imine) phase generated by excessive Na^+ ions during formation of CN-polymer structure [23]. When a larger amount of $\text{Na}_2\text{S}_2\text{O}_3$ is added (2.0 g), the product after calcination is significantly reduced. If increased to 2.4 g , there is almost no production. The reason for this phenomenon may be that excessive SO_2 gas destroys the basic structure of PCN. Fig. 2b reveals a more detailed defect structure of prepared catalysts. The characteristic peaks located in the region of $3000\text{--}3500 \text{ cm}^{-1}$, $1100\text{--}1700 \text{ cm}^{-1}$ and 810 cm^{-1} are attributed to the N–H stretching vibrations (NH_2 group), stretching vibration of the C–N heterocycle ring and bending vibration of heptazine ring, respectively [24]. It is worth noting that a shoulder

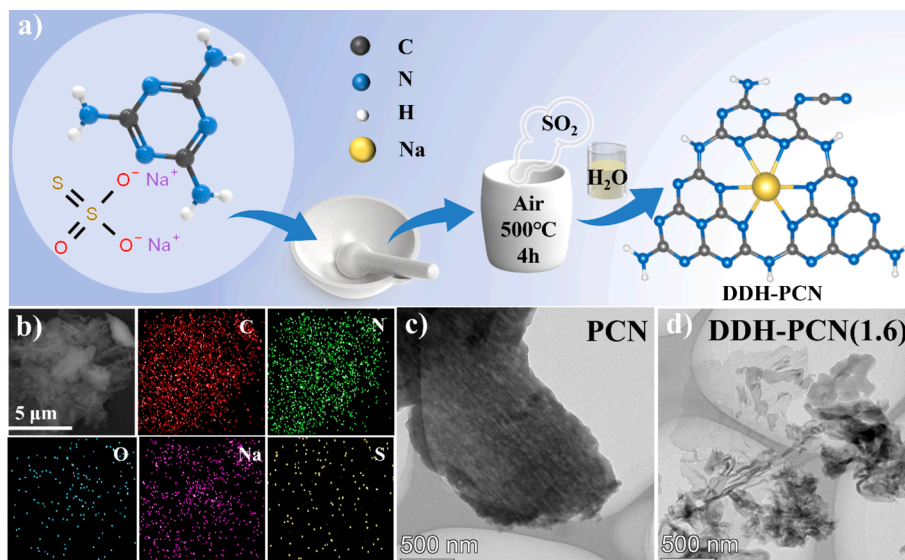


Fig. 1. (a) Schematic diagram of the preparation process of DDH-PCNs. (b) The SEM images and elemental (C, N, O, Na and S) mapping images of DDH-PCN(1.6). TEM images of (c) PCN and (d) DDH-PCN(1.6).

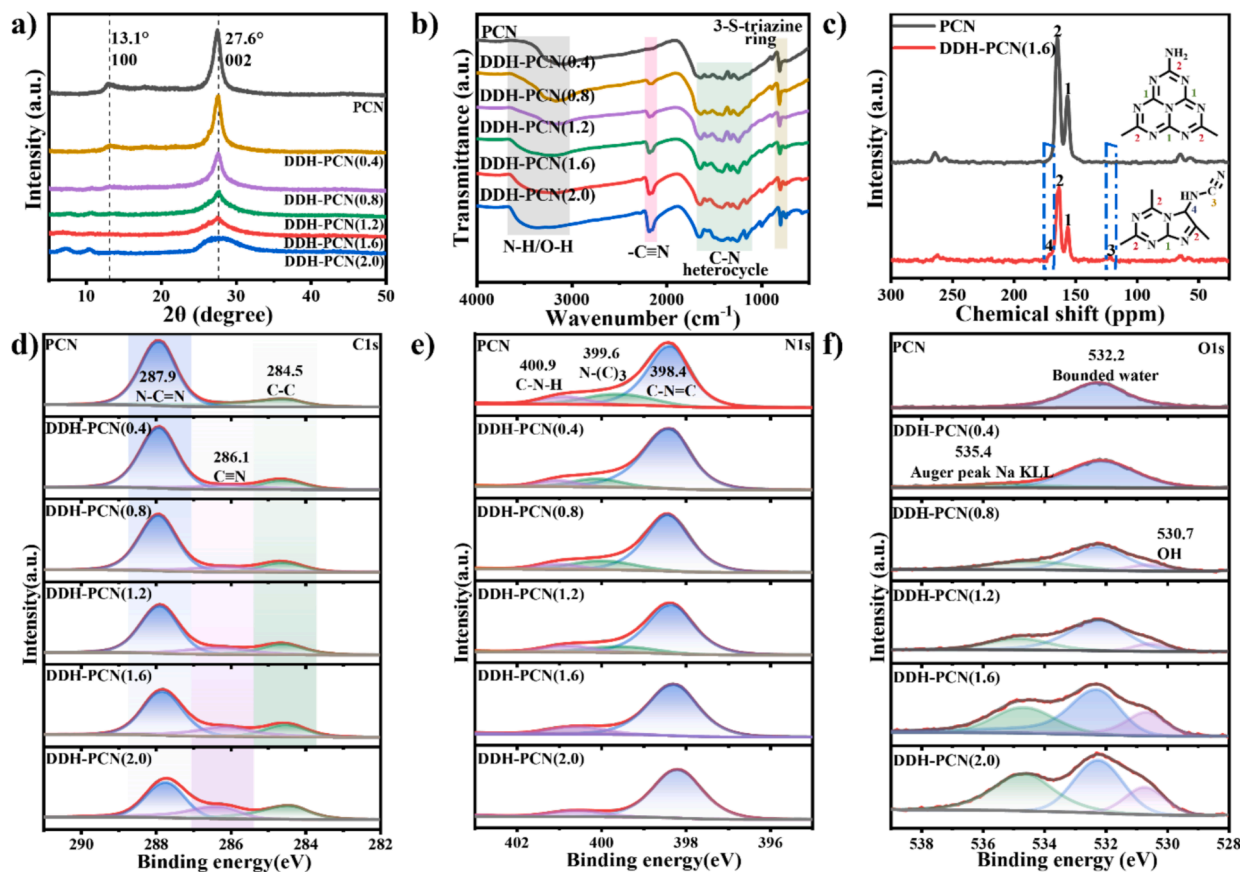


Fig. 2. (a) XRD patterns, (b) FTIR spectra, and (c) Solid-state ¹³C MAS NMR spectra of as-prepared samples. (d) C 1s, (e) N 1s and (f) O 1s XPS spectra of PCN and different DDH-PCNs.

peak belonging to the -OH group forms at around 3500 cm⁻¹ from DDH-PCN(1.2) to DDH-PCN(2.0), suggesting that the original NH₂ group gradually transformed into -OH group during acidic SO₂ gas treatment [17]. A new peak attributed to the -C≡N groups is observed at 2167 cm⁻¹ [9,25], which increasing with the addition of Na₂S₂O₃, suggesting that -C≡N is successfully introduced and its content can be

easily adjusted.

Next, the solid-state ¹³C NMR is further used to confirm the position of the -C≡N groups (Fig. 2c). The NMR spectra of PCN and DDH-PCN (1.6) show two strong peaks at 156.0 and 164.2 ppm, corresponding to C_{3N}(1) and C_{2N-NHx}(2) in the heptazine units, respectively [26,27]. The C2 peak intensity is decreased obviously, proving that the -C≡N

groups may be originated from the deprotonation of $-C-NH_2$ and located at end of the melon structure [28,29]. Moreover, the decrease in the C1/C2 peak area ratio in DDH-PCN(1.6) (Table S1) indicates a change in the surrounding environment of C and N atoms in the heptazine ring structure. There are also two new resonance peaks in DDH-PCN(1.6) at around 121.2 ppm and 170.8 ppm, which can be ascribed to carbon atom (3) in $-C\equiv N$ groups and the neighbor C atom (4), respectively, further demonstrating the formation of $-C\equiv N$ groups [30].

X-ray photoelectron spectroscopy (XPS) results display all the C1s spectrums (Fig. 2d) have two main peaks at around binding energies of 287.9 and 284.5 eV, which are assigned to the sp^2 -hybridized $N-C\equiv N$ and $C-C$, respectively [19]. A new peak at 286.1 eV ($-C\equiv N$ groups) is observed and displays increasing intensity with increasing $Na_2S_2O_3$ usage (Table S2), further demonstrating that the $-C\equiv N$ groups are formed in PCN with well-adjustability [31]. Meanwhile, the proportion of $N-C\equiv N$ peak decreases, while the intensity of $C-C$ peak increases, owing to the N atoms in $N-C\equiv N$ groups gradually vanished to form N vacancies and neighboring C atoms bonded into the $C-C$ structure, which solidly proved that the generation of N vacancies is located at $N-C\equiv N$. Similarly, the obvious decreasing of all N1s peaks intensity and the disappearance of the $N-(C)_3$ peak (399.6 eV) in DDH-PCN(1.6) can be attributed to the appearance of N vacancies (Fig. 2e). The N/C atomic ratio decreases sharply from 1.27 (PCN) to 1.07 (DDH-PCN(1.6)) (Table S3) also implies the formation of N vacancies. The similar results are observed in elemental analyzer testing (Table S4) and EPR spectra (Fig. S8). DDH-PCNs shows higher EPR intensity than PCN, indicating more unpaired electrons from defects. These unpaired electrons may come from the unpaired electrons around C atoms adjacent to the N defect. Such unpaired electrons probably provide more electrons to participate in photocatalysis [10].

In the O1s spectrum (Fig. 2f), the main peak at 532.2 eV should be owing to the bounded water. With increasing content of $Na_2S_2O_3$ dosage, a new peak at 530.7 eV appears at DDH-PCN(1.2), which can be assigned to hydroxyl group ($-OH$) [32], indicating the acidic SO_2 gas endows PCN abundant surface $-OH$ groups. The DDH-PCN(1.6) exhibits a 0° of water contact angle (Fig. S9), proving its superhydrophilic

properties. The Na^+ has also been successfully doped into PCN in the form of $(-C\equiv N)-Na^+$ with the addition of $Na_2S_2O_3$ (Fig. S10) [33]. Moreover, it is difficult to detect the S in both XPS (Table S3) and elemental analysis (Table S4), demonstrating it can be escaped in form of SO_2 gas after tailoring PCN. In general, DDH-PCN has controllable multi-configuration, which may contribute to the improvement of photocatalytic performance.

3.3. Band structure and carrier separation characteristics

UV-Vis diffuse reflectance spectroscopy (UV-Vis-DRS) is used to investigate the impact of dual-defect structures on optical response (Fig. 3a). Compared with PCN, the light harvesting of DDH-PCN samples is improved and shows a redshift in the absorption edge [28]. The increased absorbance at 350–380 nm is commonly ascribed to $\pi \rightarrow \pi^*$ transitions in co-aromatic systems, while the absorption edge redshift at 460–600 nm is caused by the electron transition of lone pair electrons $n \rightarrow \pi^*$ at the defect position ($-C\equiv N$ and N vacancies), indicating that the dual-defect sites can jointly optimize the electronic structure and enhance the absorption of visible light [34,35]. In the Tauc plots from Kubelka-Munk function transformation, the band gap values (E_g) of PCN, DDH-PCN(0.4), DDH-PCN(0.8), DDH-PCN(1.2), DDH-PCN(1.6) and DDH-PCN(2.0) are 2.83, 2.81, 2.81, 2.83, 2.77 and 2.86 eV (Fig. 3b), respectively. As determined from the VB-XPS spectra (Fig. 3c and Fig. S11), the VB maximum value of PCN and DDH-PCN(1.6) are 1.9 eV. The E_{VB} of prepared materials can be calculated using the formulas of E_{VB} (vs. NHE) = $\Phi + VB_{XPS} - 4.44$ eV, where Φ and VB_{XPS} denote the work function of the XPS analyzer (4.2 eV) and the VB potential value is tested by VB-XPS [36]. Consequently, the band structure of the DDH-PCNs are outlined in Fig. 3d. The flat band potential (E_{FB}) of all catalysts (Fig. 3e–f) shows a positive slope, indicating their n-type semiconductor properties [37]. For n-type semiconductors, the conduction band potential (E_{CB}) is usually more negative of 0.1 eV compared to flat band potential [38], thus is calculated to be -0.90 V and -0.97 V (vs.NHE) for PCN and DDH-PCN(1.6), respectively, which almost consistent with the valence band spectrum results.

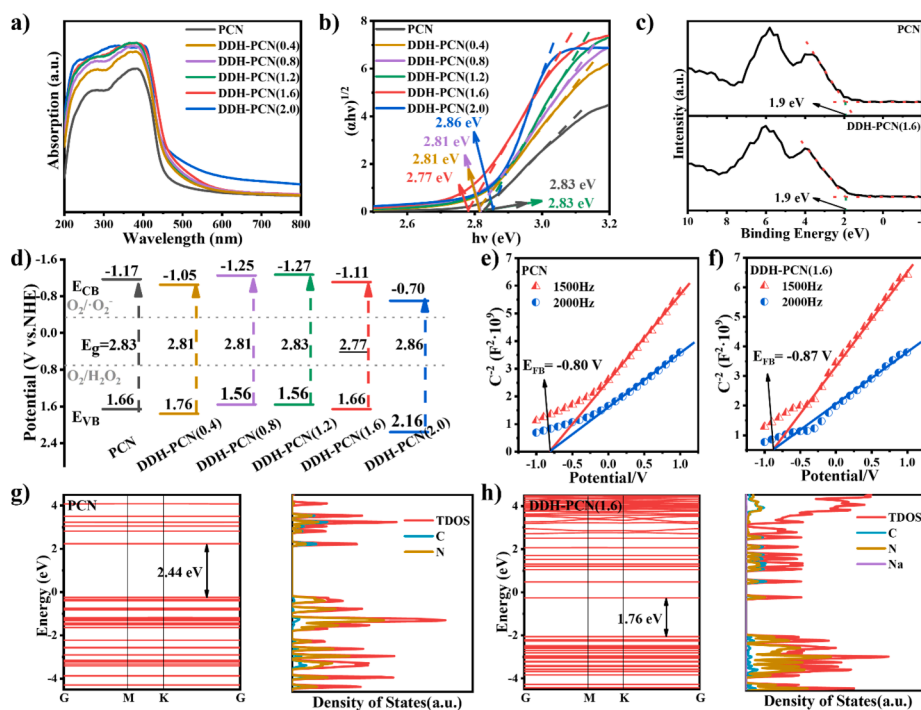


Fig. 3. (a) UV-vis DRS and (b) corresponding plots of the converted Kubelka-Munk function versus photon energy for PCN and DDH-PCNs. (c) XPS valence and (d) the band structure of as-prepared samples. (e–f) Mott-Schottky (M–S) curves. (g–h) Calculated band structure and projected density of states (TDOS) for PCN and DDH-PCN(1.6).

Density functional theory (DFT) calculation is carried out to explore the influence of dual defects on band structure and electronic structure (Fig. 3g–h). The calculated band results indicate that VB is mainly formed by N 2p orbitals, while CB originates from C 2p and N 2p orbitals [39]. New orbital energy levels appeared in the DDH-PCN(1.6), and the band gap decreased from 2.44 eV in PCN to 1.76 eV. Although the band gap energy of the sample calculated by DFT is lower than the experimental value due to the well-known limitations of DFT calculation, the narrowed band gap of DDH-PCN(1.6) is consistent with the previous results of DRS. The above results indicate that the synergistic effect of $\text{—C}\equiv\text{N}$ groups and N vacancies can significantly reduce the band gap and promote light absorption.

The influence of dual-defect sites on the separation and transport of photogenerated charge carriers is also studied through TPR, EIS, PL and TRPL. Fig. 4a shows that both PCN and DDH-PCN(1.6) exhibited a sensitive photocurrent response during visible light on/off cycles. DDH-PCN(1.6) possesses almost 3 times of photocurrent density higher than PCN, indicating that dual defects can more effectively achieve photo-excited charge separation. Moreover, DDH-PCN(1.6) has a smaller arc radius than PCN in EIS spectra (Fig.S12), meaning that DDH-PCN(1.6) has lower interface electron transfer resistance [40]. PL and TRPL spectrum were employed to investigate the migration, transfer, and recombination processes of photogenerated electron-hole pairs in semiconductors [40,41]. Compared with PCN, the PL intensity for DDH-PCN(1.6) is significantly reduced (Fig.S13) [42]. With the amount of $\text{Na}_2\text{S}_2\text{O}_3$ increasing, the strongest emission peak displayed a blue shift, which is due to quantum confinement caused by a decrease in size and thickness [43,44]. DDH-PCN(1.6) also exhibits faster TRPL decay kinetics (average lifetime = 2.97 ns) than PCN (3.88 ns) (Fig. 4b). Meanwhile, the non-symmetrical electrostatic potential (Fig. 4c) of PCN and DDH-PCN(1.6) reflects the uneven distribution of charge density, which will cause the dipole of molecules, promoting the effective exciton dissociation. The highest occupied and lowest unoccupied molecular orbitals (HOMO and LUMO) refer to the bonding orbitals and anti-bonding orbitals, respectively, which are related to the hole and electron distribution upon photoexcitation. For PCN, LUMO and HOMO are uniformly distributed on each heptazine ring (Fig. 4d). Such fully overlap resulting in the highest recombination possibility followed by light irradiation. In stark contrast to PCN, the electron density is redistributed in DDH-PCN(1.6), forming an electron rich region (Fig. 4e). The electrons of LUMO are mainly distributed on $\text{—C}\equiv\text{N}$ groups and N vacancies, while the electrons of HOMO gathered on the other side. The special orbital redistribution confirms that introducing $\text{—C}\equiv\text{N}$ groups and N vacancies can generate a new path of electron transfer, promoting the separation of electrons and holes and effectively suppressing

recombination.

3.4. Photocatalytic performance of H_2O_2 generation

The photocatalytic production efficiency of H_2O_2 by different samples under visible light and air conditions is evaluated (Fig. 5a–b). The H_2O_2 yield of DDH-PCN(1.6) reaches 36.7 μM within 30 min with generation rate of 1713.1 $\mu\text{mol g}^{-1} \text{h}^{-1}$, which is 50 times higher than PCN (35.1 $\mu\text{mol g}^{-1} \text{h}^{-1}$). Interestingly, a sudden jump in yield from DDH-PCN(1.2) to DDH-PCN(1.6) is due to the presence of a deprotonated hydroxyl group (Fig. 2f), which improves the hydrophilicity of the catalyst. Notably, the production of H_2O_2 will increase to 5554.6 $\mu\text{mol g}^{-1} \text{h}^{-1}$ under full spectrum light irradiation (Fig. 5c), further verifying the outstanding catalytic activity of DDH-PCN(1.6). To investigate the long-term activity, Fig. 5d compares the performance of PCN and DDH-PCN(1.6) in producing H_2O_2 during 6 h. The H_2O_2 production yield increases with the extension of time, in which the DDH-PCN(1.6) is superior and the H_2O_2 concentration reached to 424.8 μM after 6 h. Meanwhile, the decomposition efficiency of DDH-PCN(1.6) for H_2O_2 is extremely low (Fig.S14). The apparent quantum yield (AQY) under the monochromatic light illumination is measured, where the DDH-PCN (1.6) achieved 2.8 % at 400 nm (Fig. 5e). It also shows good stability in photocatalytic cycle experiments (Fig.S15). The XRD, FTIR and XPS spectras also exhibit negligible variation after cyclic experiments (Fig. S16). Finally, compared with reported photocatalysts recently, DDH-PCN(1.6) has certain advantages in photogeneration of H_2O_2 (Table S5), proving the key role of dual defects in enhancing photocatalytic ORR activity.

Fig.S17 shows that the ORR cannot occur under dark conditions or without photocatalysts. In addition, the yield of H_2O_2 is almost completely inhibited when adding AgNO_3 or bubbling N_2 . These results prove that the H_2O_2 production is a photo-driven O_2 reduction reaction. Actually, the photocatalytic production of H_2O_2 can be achieved through two pathways: a continuous two-step single electron (Eq. (4)–(5)) or a one-step double electron reduction reaction (Eq. (6) [45–47]. After adding p-benzoquinone (pBQ) as $\cdot\text{O}_2^-$ capture agents, almost no H_2O_2 generated (Fig.S17). The existence of $\cdot\text{O}_2^-$ is further confirmed through EPR by using 5,5-dimethyl-1-pyrroline N-oxide (DMPO) (Fig. 5f), in which DDH-PCN(1.6) exhibits a higher signal intensity than that of PCN, indicating that the synergistic effect of dual defects helps activate O_2 to $\cdot\text{O}_2^-$, thereby accelerating the production of H_2O_2 [48].

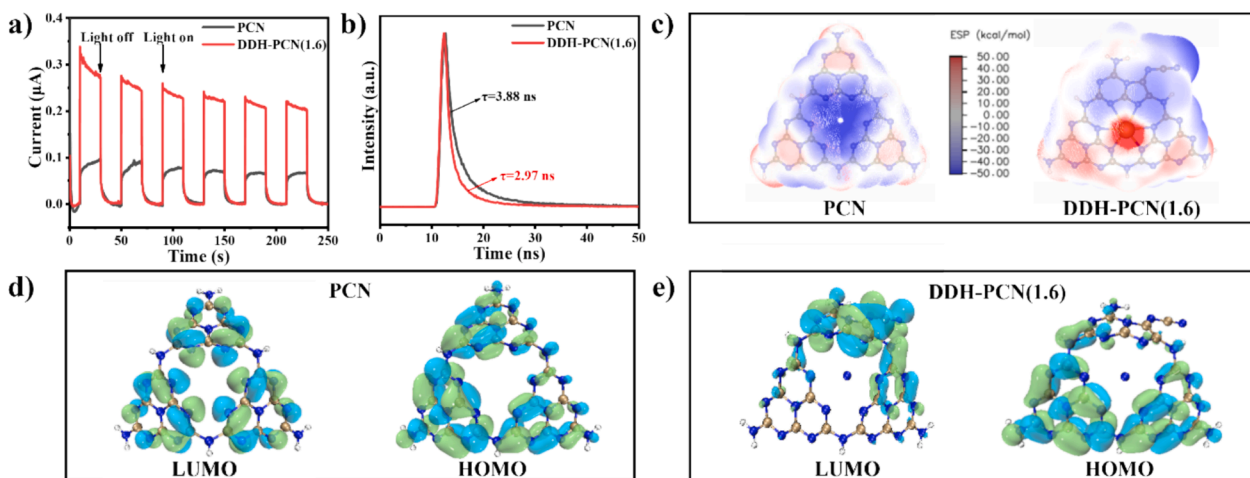


Fig. 4. (a) Transient photocurrent response. (b) Time-resolved PL spectra (TRPL). (c) The electrostatic potential surface distribution. (d-e) Highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) distribution of PCN and DDH-PCN(1.6).

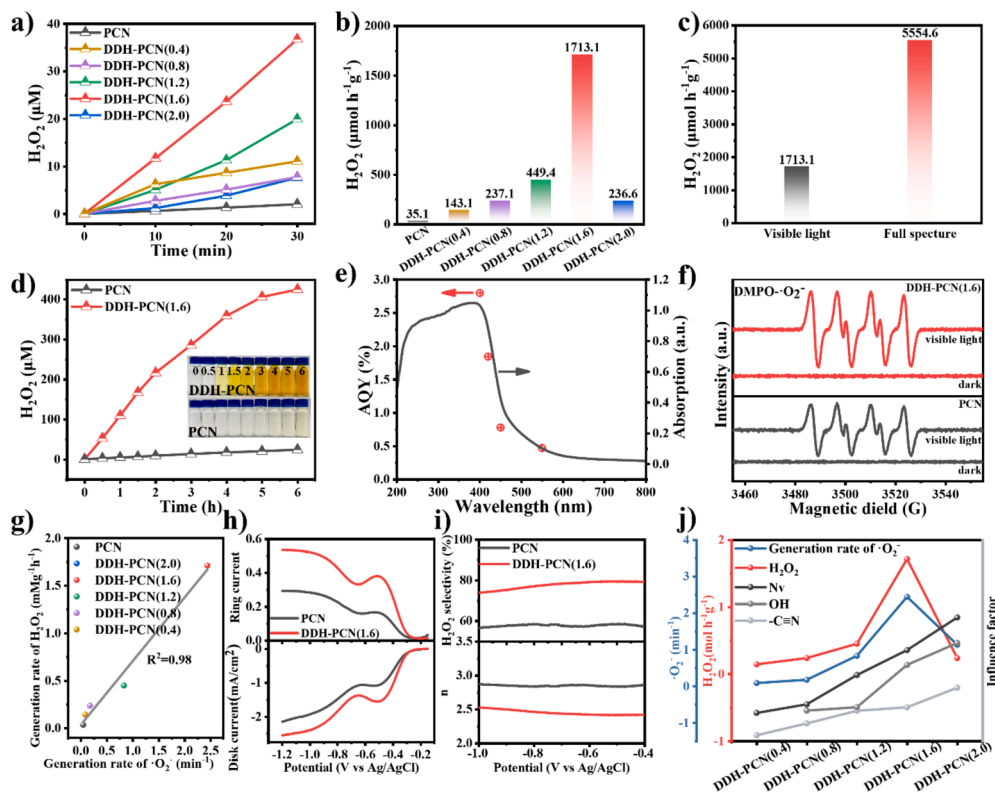


Fig. 5. (a) Photocatalytic activities of H_2O_2 production and (b) corresponding evolution rate over different samples. (c) Photocatalytic performance of DDH-PCN(1.6) under full spectrum and visible light irradiation. (d) Time-dependence of photocatalytic H_2O_2 production. (e) The wavelength-dependent AQY. (f) EPR spectra. (g) Dependence of the H_2O_2 evolution rate on the $\cdot\text{O}_2^-$ intermediate. (h) RRDE polarization curves. (i) Selectivity of H_2O_2 (upper) and corresponding average number of transferred electrons (lower). (j) The relationship between H_2O_2 yield, rate- O_2^- generation and various configuration.



To quantify the $\cdot\text{O}_2^-$ production rate over DDH-PCN(1.6) photocatalytic system, NBT capture experiments have been conducted. The NBT can react specifically with $\cdot\text{O}_2^-$ to generate blue purple complex precipitates, which can be detected by its absorbance at 259 nm through UV-vis spectrophotometer. Fig.S18 intuitively reflects that DDH-PCN(1.6) consumes the largest amount of NBT within 15 min, which means it has the fastest $\cdot\text{O}_2^-$ production efficiency than other photocatalysts. The kinetics of $\cdot\text{O}_2^-$ generation can be calculated using the formula $k = dC / dt$ (Fig.S19), where k represents the rate of $\cdot\text{O}_2^-$ generation, dt represents the reaction time, and dC represents the amount of $\cdot\text{O}_2^-$ generated during the reaction time (corresponding to the decrease in absorbance of the UV-vis spectrophotometer at 259 nm). According to calculations, the $\cdot\text{O}_2^-$ generating rate of DDH-PCN(1.6) up to 2.44 min^{-1} , which is almost 60 times faster than that of PCN ($4.34 \times 10^{-2} \text{ min}^{-1}$). We simultaneously fit the rate of $\cdot\text{O}_2^-$ production and H_2O_2 production by different catalysts (Fig. 5g), which shows a linear relationship, further confirming that H_2O_2 is generated by two-step single electron reduction using $\cdot\text{O}_2^-$ as the intermediate.

The influence of dual-defect sites in DDH-PCN(1.6) on ORR selectivity is investigated through voltammetry of the rotating ringdisk electrodes (RRDE). The disk current comes from the reduction reaction of O_2 , and the negative reduction current is measured. Meanwhile, H_2O_2 generated on the disk electrode rapidly diffuses to the Pt electrode and is oxidized, detecting a positive oxidation electrode [49]. In the disk current of PCN and DDH-PCN(1.6) (Fig. 5h, below), a two-step ORR can be observed [50]. The sharp decrease at -0.3 V is due to the reduction of O_2 to H_2O_2 , and the oxidation current from the ring electrode increases from approximately -0.3 V to -0.5 V . PCN and DDH-PCN(1.6) exhibit similar ring current trends (Fig. 5h, above), but the ring current of DDH-

PCN(1.6) is significantly higher than that of PCN, indicating that DDH-PCN(1.6) has better oxygen reduction activity [51]. Fig. 5i shows the transfer electron number of DDH-PCN(1.6) is closer to 2, and the H_2O_2 selectivity reaches $\sim 80\%$. These results indicate that DDH-PCN(1.6) with dual defects has higher selectivity towards H_2O_2 and is more inclined to continuous two-step single electron ORR.

Finally, the main reasons for the improved performance of photocatalytic synthesis of H_2O_2 in the DDH-PCN system are discussed in detail. We compare several anions and found that $\text{S}_2\text{O}_3^{2-} \gg \text{SO}_4^{2-} > \text{Cl}^-$ in terms of photocatalytic activity, indicating that the key effect of $\text{S}_2\text{O}_3^{2-}$ in photocatalyst fabrication (Fig.S20). Next, the relationship between H_2O_2 yield, generation rate of $\cdot\text{O}_2^-$ and dual defects content is plotted (Fig. 5j). The PCN-NaCl and PCN-KCl were prepared by using NaCl or KCl, respectively, which only contains $-\text{C}\equiv\text{N}$ groups. The PCN-NaCl and PCN-KCl shows the H_2O_2 yield of 66.4 and $149.7 \mu\text{mol g}^{-1} \text{ h}^{-1}$ (Fig.S20), respectively, which is significantly lower than that of DDH-PCN, ruling out the key effect of $-\text{C}\equiv\text{N}$ groups on improving performance. However, there is a special orbital redistribution in $-\text{C}\equiv\text{N}$ -PCN even no Nv, suggesting that the $-\text{C}\equiv\text{N}$ groups can provide the asymmetric electron distribution for carriers' separation (Fig.S21). For Nv, the suitable concentration will positively influence on photocatalytic activity. Its effect on photocatalytic ORR performance is mainly focus on the optimization of O_2 adsorption and activation (the detail see the analysis in Fig. 6). When the hydroxyl group content quickly increases, a jump of photocatalytic H_2O_2 generation rate was observed in DDH-PCN(1.6) (Fig. 5b and Fig. 5j), implying that a positive correlation between hydroxyl group and photocatalytic H_2O_2 production. Moreover, as shown in Fig.S16f, the peak of surficial hydroxyl groups was obviously decreased after ORR pathway, probably indicating it can provide protons by consuming itself [52]. Finally, the trend of superoxide radical yield is well consistent with that of H_2O_2 yield, indicating that the $\cdot\text{O}_2^-$ intermediate is the key to photo-synthesis of H_2O_2 .

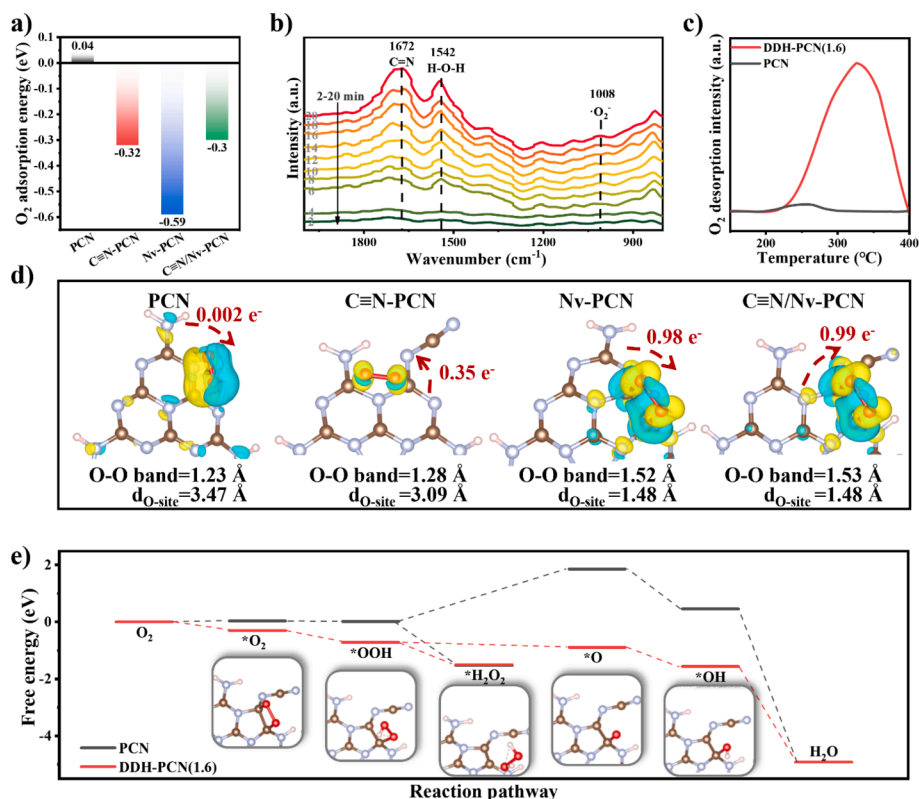


Fig. 6. (a) O_2 adsorption energy. (b) *In-situ* DRIFT spectra during the photocatalytic H_2O_2 production of DDH-PCN(1.6) (water system). (c) Temperature programmed oxygen desorption (O_2 -TPD) profiles. (d) The charge density difference between models with different defects adsorbing O_2 (yellow represents the area of electron accumulation, light blue represents the area of electron dissipation). (e) Free energy profiles for photocatalytic H_2O_2 evolution reactions.

3.5. Mechanism of photocatalytic H_2O_2 evolution

In order to gain a deeper understanding of the role and catalytic activity of each defect site in the $2e^-$ ORR, PCN models with different defect structures are constructed (Fig.S22), and density functional theory (DFT) calculations are performed to explore the basic ORR steps. Generally, the $2e^-$ ORR steps include: (i) O_2 adsorption on the catalyst surface ($*O_2$), (ii) $*O_2$ gains an electron and combines with $*H$ to form $*OOH$, and (iii) $*OOH$ captures another electron and $*H$ to generate H_2O_2 . Firstly, the O_2 adsorption energy (E_{ads}) of PCN is +0.04 eV, indicating that it is difficult to adsorb O_2 on pure PCN surface (Fig. 6a). However, N vacancies can efficiently capture O_2 through a bridge adsorption model (C–O–O–C) with the lowest E_{ads} value of –0.59 eV. Further introduction of C≡N group, C≡N/Nv-PCN still exhibits a negative E_{ads} value of –0.30 eV, illustrating it also can thermodynamically spontaneously adsorb O_2 molecules. *In-situ* diffuse reflection infrared Fourier transform spectroscopy (DRIFTS) (Fig. 6b and Fig.S23) shows that DDH-PCN has two vibrational peaks at 1008 cm^{-1} ($\cdot O_2^-$ adsorption state) and 898 cm^{-1} (peroxy species, $-O-O-$) [53]. In particular, with increasing of the O_2 adsorption, the bands of 1672 cm^{-1} (C=N) and 1542 cm^{-1} (H–O–H) change greatly, which can be attributed to the change of dipole moment caused by electron transfer after oxygen adsorption [54]. In addition, gas chromatography (GC) was also used to investigate the decomposition products of ethanol during the reaction (Fig.S24). We simultaneously detected acetaldehyde and acetic acid in DDH-PCN reaction system. This may be because DDH-PCN(1.6) can quickly adsorb O_2 and generate $\cdot O_2^-$, which then immediately combine with the proton provided by ethanol to form H_2O_2 and product of acetaldehyde/acetic acid.

In O_2 -TPD spectra, the area of the desorption peak represents the amount of O_2 that can be adsorbed per unit mass of catalyst, and the desorption temperature represents the strength of surface interaction of

O_2 . As shown in Fig. 6c, the surface of DDH-PCN(1.6) exhibits higher O_2 adsorption capacity. However, its BET is significantly lower than that of PCN (Fig.S25 and Table S6), which means that DDH-PCN(1.6) exhibits much higher O_2 adsorption density on the surface.

The Bader charge and stretching of O–O bond accurately reflected the activation of oxygen (Fig. 6d). The PCN shows almost no change of electronic density and no apparent O–O bond stretching, suggesting that O_2 molecules are difficult to be activated on the PCN surface. However, the PCN with N vacancies can transfer the average of $0.98 e^-$ to adsorbed O_2 , while PCN with C≡N will efficiently elongate the O–O bond. Both of these two virtues converge on C≡N/Nv-PCN, showing the most electron transfer and the longest O–O bond length, which testifies the synergistic effect of C≡N and N vacancies. Furthermore, the introduction of N vacancies and C≡N groups resulted in a close distance between O_2 and PCN, decreasing from the initial 3.47 Å to 1.48 Å (Fig.S22). In the process of $*OOH$ formation (Fig. 6e), a key intermediate for the subsequent formation of H_2O_2 , the ΔG value of DDH-PCN(1.6) is –0.42 eV, much lower than the –0.02 eV of PCN. Moreover, the $*OOH$ is more inclined to transform into H_2O_2 rather than H_2O , suggesting its good selectivity.

3.6. Photocatalytic degradation of SMX

Except for the photocatalytic production of H_2O_2 , the DDH-PCNs have also been proven to show excellent photocatalytic properties of TrOCs degradation. As shown in Fig. 7a–b, the degradation rate of SMX (1 mg/L) by PCN is $1.9 \times 10^{-2}\text{ min}^{-1}$, while DDH-PCN(1.6) reaches $1.8 \times 10^{-1}\text{ min}^{-1}$. Moreover, the degradation rate of DDH-PCN(2.0) is further increased to $3.4 \times 10^{-1}\text{ min}^{-1}$, which is 18 times that of PCN. It is probably due to the rich surface hydroxyl and deprotonated hydroxyl groups can enhance the contact with SMX contaminants. On the contrary, the degradation effect of PCN is better than that of samples with

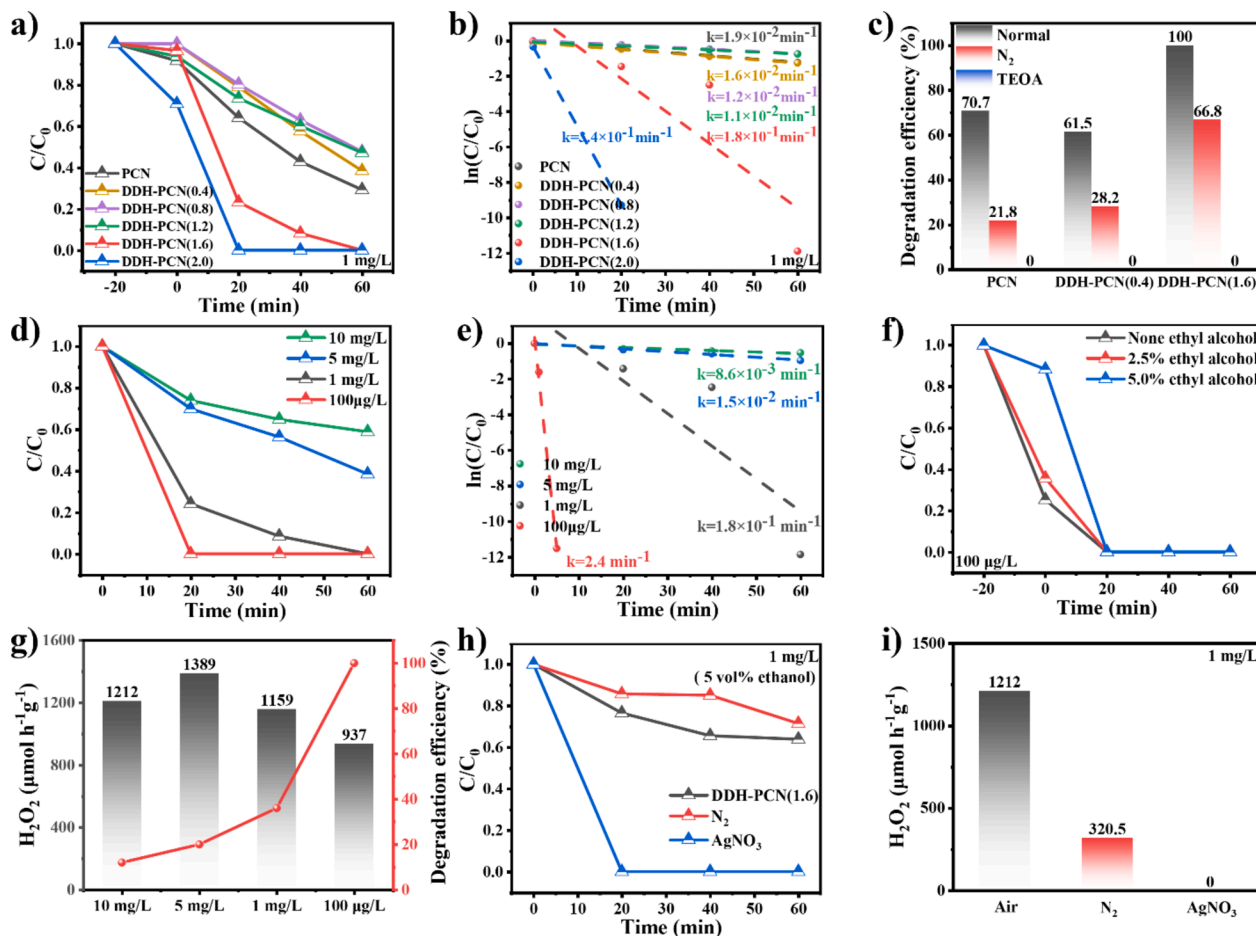


Fig. 7. (a–b) The degradation efficiency and corresponding rate of SMX (1 mg/L) over different samples. (c) The degradation efficiency of SMX by PCN, DDH-PCN (0.4) and DDH-PCN(1.6) in the absence of $\cdot\text{O}_2^-$ and holes. (d–e) The effect of different SMX concentrations and (f) ethanol concentrations on photocatalytic degradation. (g) Simultaneous photocatalytic degradation of SMX and H_2O_2 production over DDH-PCN(1.6) in ethanol (5 %)/wastewater system, (h–i) and the corresponding sacrificial agent experiment.

low concentration defects, such as DDH-PCN(0.4), DDH-PCN(0.8), and DDH-PCN(1.2), probably because the defects in low concentration act as the recombination center of electron and holes. A free radical scavenging experiment (Fig. 7c) shows that both $\cdot\text{O}_2^-$ and photogenerated h^+ are factors affecting degradation.

The degradation experiments with different SMX concentrations are further conducted over DDH-PCN(1.6) (Fig. 7d–e). Under the high concentration of SMX, both PCN and DDH-PCN(1.6) show the low degradation efficiency (Fig.S26), but the degradation efficiency of DDH-PCN(1.6) is still higher than PCN. The degradation rate will be further increased to 2.4 min^{-1} when SMX concentration reduce into $100 \mu\text{g/L}$. These results indicate that the good dispersion of DDH-PCN may provide more accessible sites, favoring the capture and degradation of TrOCs.

Interestingly, we found that the DDH-PCN(1.6) can simultaneously achieve photocatalytic H_2O_2 production and SMX degradation in a wastewater system containing alcohol (Fig. 7f–g). DDH-PCN(1.6) can degrade $100 \mu\text{g/L}$ of SMX (5 vol% of ethanol) in 20 min with a rate of $4.6 \times 10^{-1} \text{ min}^{-1}$, while the amount of H_2O_2 yield is $937 \mu\text{mol g}^{-1} \text{ h}^{-1}$. The generation of H_2O_2 is also directly proportional to the ethanol content, which coincides with our previous discussion (Fig.S27). Even in presence of ethanol, trace organic pollutants can be completely degraded, proving that the strong oxidation ability of $\cdot\text{O}_2^-$ of DDH-PCN. Free radical capture experiments (Fig. 7h) displayed that bubbling N_2 will obviously inhibit both degradation and H_2O_2 production process, proving the $\cdot\text{O}_2^-$ is the key active species in wastewater/alcohol system. However, a small amount of H_2O_2 can still be detected with the yield of

$320.5 \mu\text{mol g}^{-1} \text{ h}^{-1}$ (Fig. 7i) under the N_2 atmosphere, which probably due to the oxidation of H_2O by photogenerated holes. On the contrary, we find that the degradation effect of SMX increased linearly after addition of AgNO_3 , because Ag^+ can capture photogenerated electrons and deposit them on the catalyst surface, accelerating separation of photogenerated electrons and hole pairs. In addition, DDH-PCN(1.6) also exhibits good removal performance in RhB wastewater/alcohol system (Fig.S28). From these results, it can be concluded that the dual-defect catalyst is fully rely on $\cdot\text{O}_2^-$ to achieve both photocatalytic H_2O_2 production and TrOCs degradation in wastewater/alcohol system.

In general, the highest occupied molecular orbital (HOMO) are more susceptible to attacks from electrophilic free radicals [55,56], while the lowest vacant molecular orbital (LUMO) represents electron acceptance ability and tends to nucleophilic attacks [57,58]. From Fig. 8a, the HOMO orbitals of SMX primarily reside on the benzene ring, encompassing the C atom on the benzene ring and the adjacent N (17) atom. Conversely, the LUMO orbitals concentrate around atoms with high electronegativity, such as S8 and N7. Fukui functions (f^- , f^+ , f^0) are used to further explore the potential activation of atoms within the SMX structure (Fig. 8b). The regions with high Fukui functions (f^0) are represented as reaction sites in AOPs that are attacked by free radicals [59]. Specifically, the N5 ($f^0 = 0.0661$), C14 ($f^0 = 0.0622$), and N17 ($f^0 = 0.0798$) exhibit higher f^0 values, indicating that they are the main sites for free radical attacks on SMX molecules.

Next, mass spectrometry (MS) analysis is performed on the intermediate products of SMX degradation [60] (Fig. 8c). Under the action of

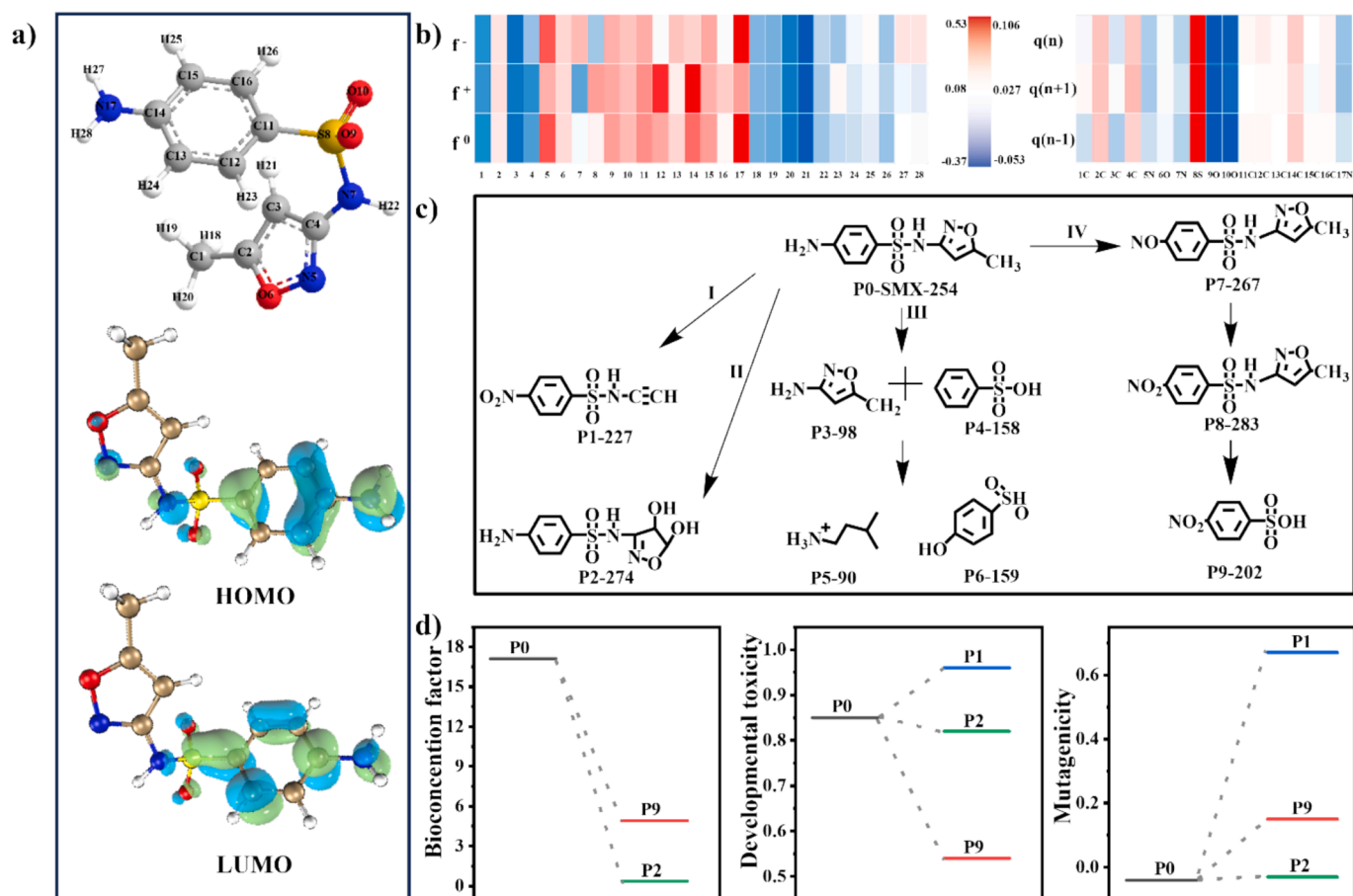


Fig. 8. (a) Optimized molecular structure, HOMO and LUMO of SMX. (b) Calculated Fukui indexes. (c) Possible degradation pathways. (d) Toxicology predicted by T.E.S.T. (P0 refers to the ionized state of SMX).

ring opening and coupling reactions, the amino group (N17 atom) on the benzene ring is oxidized (path I). Meanwhile, the addition of C–C bonds and the opening of isoxazole ring (N5 and O6) is occurred, which leads to the product P1 ($m/z = 227$). The double bond on isoxazole ring in pathway II is attacked by electrophilic and oxidized to P2 ($m/z = 274$) [61]. In pathway III, the S–N bond of SMX is easily attacked by free radicals to form P3 ($m/z = 98$) and P4 ($m/z = 158$), and then forming P5 ($m/z = 90$) [62] through cleavage of the isoxazole ring, while hydroxyl substitution is converted to P6 ($m/z = 159$). The intermediate product of 4-nitroso-SMX (P7 ($m/z = 267$)) was found in pathway IV, which could be attributed to the oxidation of amino group (N17 atom) in the benzene ring. The further oxidation of 4-nitroso-SMX resulted in the formation of 4-nitro-SMX (P8 ($m/z = 283$)) and then 4-nitrobenzenesulfonic acid (P9 ($m/z = 202$)). This means that the free radical can attack the isoxazole group of SMX and produce products similar to SMX's molecular formula.

The consensus method in the Toxicity Estimation Software Tool (T.E.S.T.) is used to evaluate the toxicity of degradation intermediates in the photocatalytic degradation of SMX system. As shown in Fig. 8d, P2 and P9 showed a significant downward trend in bioconcentration factor and developmental toxicity, indicating that the cumulative toxicity of SMX to organisms is weakened. In addition, both by-products P2 and P9 showed negative mutagenicity (mutagenicity < 0.5), indicating reduced toxicity. Therefore, due to the attenuation effect of comprehensive toxicity, the proposed degradation of TrOCs by DDH-PCNs is feasible and environmentally friendly.

4. Conclusion

In summary, dual-defect PCN with hydroxyl groups (DDH-PCNs) was

constructed through one-step acid gas tailoring strategy. The key role of the various configuration in the photocatalytic reaction was clearly clarified: (i) The $-\text{C}\equiv\text{N}$ groups provide the asymmetric electron distribution for carriers' separation. (ii) The Nv accelerate the O_2 adsorption and activation by formation of $\text{C}-\text{O}-\text{O}-\text{C}$ bridge model. (iii) The surficial hydroxyl will provide more protons for photocatalytic H_2O_2 generation. As a result, the photocatalytic H_2O_2 rate of DDH-PCNs is up to $5554.6 \mu\text{mol g}^{-1} \text{h}^{-1}$ under full spectrum, which is 30 times higher than PCN. DDH-PCNs can also completely degrade SMX ($100 \mu\text{g/L}$) within 20 min with a rate of $3.3 \times 10^{-1} \text{ min}^{-1}$. More importantly, the optimal DDH-PCN(1.6) can simultaneously achieve photocatalytic H_2O_2 production and SMX degradation in a wastewater/alcohol system, which is promising for the treatment of alcohol-containing organic wastewater. This study provides the profound insights into development of dual functional photocatalytic system towards environmental application.

CRediT authorship contribution statement

Na Li: Conceptualization, Methodology, Writing – original draft. **Chen Wang:** Data curation, Wei Gao: Investigation, Visualization. **Wenli Li:** Investigation. **Binrong Li:** Writing – review & editing, Supervision. **Xuedong Wang:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research is financially supported by National Natural Science Foundation of China (No. 22306142 and 22076134); Natural Science Foundation of Jiangsu Province (BK20230656, BK20211338); National & Local Joint Engineering Laboratory for Municipal Sewage Resource Utilization Technology, Suzhou University of Science and Technology (No. 2022KF02); Key Science & Technology Project of Suzhou City (SS202028) and Jiangsu Provincial Key Research & Development Program (BE2022733). The authors would like to thank Yankai Li from Shiyanjia Lab (www.shiyanjia.com) for the electrochemistry test. In addition, the DFT calculation was supported by Bianshui Riverside Supercomputing Center (BRSC).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.seppur.2024.129978>.

References

- [1] Y. Shiraiishi, Y. Ueda, A. Soramoto, S. Hinokuma, T. Hirai, Photocatalytic hydrogen peroxide splitting on metal-free powders assisted by phosphoric acid as a stabilizer, *Nat. Commun.* 11 (2020) 3386.
- [2] J. Ma, N.A. Choudhury, Y. Sahai, A comprehensive review of direct borohydride fuel cells, *Renew. Sustain. Energy Rev.* 14 (2010) 183–199.
- [3] Z. Teng, Q. Zhang, H. Yang, K. Kato, W. Yang, Y.-R. Lu, S. Liu, C. Wang, A. Yamakata, C. Su, B. Liu, T. Ohno, Atomically dispersed antimony on carbon nitride for the artificial photosynthesis of hydrogen peroxide, *Nat. Catal.* 4 (2021) 374–384.
- [4] Y. Yi, L. Wang, G. Li, H. Guo, A review on research progress in the direct synthesis of hydrogen peroxide from hydrogen and oxygen: noble-metal catalytic method, fuel-cell method and plasma method, *Catal. Sci. & Technol.* 6 (2016) 1593–1610.
- [5] M.-G. Seo, H.J. Kim, S.S. Han, K.-Y. Lee, Direct synthesis of hydrogen peroxide from hydrogen and oxygen using tailored Pd nanocatalysts: A review of recent findings, *Catal. Surveys from Asia* 21 (2017) 1–12.
- [6] P. Song, J. Du, Y. Shi, X. Fang, X. Ma, D. Liu, D. Cao, J. Li, R. Chen, C. Wu, J. Cui, Y. Wang, J. Di, J. Low, T. Kong, R. Long, H. Zhang, Y. Xiong, Highly active 3D composites for a flow-through photocatalytic membrane reactor toward water micropollutant removal, *ACS Mater. Lett.* 6 (2024) 427–437.
- [7] M.B. Asif, J. Hou, W.E. Price, V. Chen, F.I. Hai, Removal of trace organic contaminants by enzymatic membrane bioreactors: Role of membrane retention and biodegradation, *J. Memb. Sci.* 611 (2020) 118345.
- [8] J.T. Alexander, F.I. Hai, T.M. Al-aboud, Chemical coagulation-based processes for trace organic contaminant removal: Current state and future potential, *J. Environ. Manage.* 111 (2012) 195–207.
- [9] Z. Teng, H. Yang, Q. Zhang, W. Cai, Y.R. Lu, K. Kato, Z. Zhang, J. Ding, H. Sun, S. Liu, C. Wang, P. Chen, A. Yamakata, T.S. Chan, C. Su, T. Ohno, B. Liu, Atomically dispersed low-valent Au boosts photocatalytic hydroxyl radical production, *Nat. Chem.* 16 (2024) 1250–1260.
- [10] L.-L. Liu, F. Chen, J.-H. Wu, J.-J. Chen, H.-Q. Yu, Synergy of crystallinity modulation and intercalation engineering in carbon nitride for boosted H₂O₂ photosynthesis, *Proc. Natl. Acad. Sci.* 120 (2023) e2215305120.
- [11] Q. Liang, Z. Li, Z.-H. Huang, F. Kang, Q.-H. Yang, Holey graphitic carbon nitride nanosheets with carbon vacancies for highly improved photocatalytic hydrogen production, *Adv. Funct. Mater.* 25 (2015) 6885–6892.
- [12] Y. Kang, Y. Yang, L.-C. Yin, X. Kang, G. Liu, H.-M. Cheng, An amorphous carbon nitride photocatalyst with greatly extended visible-light-responsive range for photocatalytic hydrogen generation, *Adv. Mater.* 27 (2015) 4572–4577.
- [13] Y. Kang, Y. Yang, L.-C. Yin, X. Kang, L. Wang, G. Liu, H.-M. Cheng, Selective breaking of hydrogen bonds of layered carbon nitride for visible light photocatalysis, *Adv. Mater.* 28 (2016) 6471–6477.
- [14] X. Zhang, P. Ma, C. Wang, L. Gan, X. Chen, P. Zhang, Y. Wang, H. Li, L. Wang, X. Zhou, K. Zheng, Unraveling the dual defect sites in graphitic carbon nitride for ultra-high photocatalytic H₂O₂ evolution, *Energy. Environ.* 15 (2022) 830–842.
- [15] Y. Ma, H. Sun, Q. Wang, L. Sun, Z. Liu, Y. Xie, Q. Zhang, C. Su, D. Fan, Driving hydrogen peroxide artificial photosynthesis and utilization for emerging contaminants removal by cyanided polymeric carbon nitride, *Appl. Catal. B: Environ. Energy* 335 (2023) 122878.
- [16] B. Li, N. Li, Z. Guo, C. Wang, Z. Zhu, X. Tang, G. Andrew Penyikie, X. Wang, P. Huo, Understanding the effect of multiple configurations in polymeric carbon nitride for efficient solar-driven hydrogen peroxide photosynthesis, *Chem. Eng. J.* 491 (2024) 152022.
- [17] Q. Chen, C. Lu, B. Ping, G. Li, J. Chen, Z. Sun, Y. Zhang, Q. Ruan, L. Tao, A hydroxyl-induced carbon nitride homojunction with functional surface for efficient photocatalytic production of H₂O₂, *Appl. Catal. B: Environ. Energy* 324 (2023) 122216.
- [18] P. Xia, B. Zhu, J. Yu, S. Cao, M. Jaroniec, Ultra-thin nanosheet assemblies of graphitic carbon nitride for enhanced photocatalytic CO₂ reduction, *J. Mater. Chem. A* 5 (2017) 3230–3238.
- [19] B. Li, Z. Guo, Y. Feng, M. Meng, Y. Pan, Y. Zhang, Boosting photosynthetic H₂O₂ of polymeric carbon nitride by layer configuration regulation and fluoride-potassium double-site modification, *ACS Appl. Mater. Inter.* 14 (2022) 43328–43338.
- [20] C. Feng, L. Tang, Y. Deng, J. Wang, J. Luo, Y. Liu, X. Ouyang, H. Yang, J. Yu, J. Wang, Synthesis of leaf-vein-like g-C₃N₄ with tunable band structures and charge transfer properties for selective photocatalytic H₂O₂ evolution, *Adv. Funct. Mater.* 30 (2020) 2001922.
- [21] Y. Zhang, Q. Cao, A. Meng, X. Wu, Y. Xiao, C. Su, Q. Zhang, Molecular heptazine-triazine junction over carbon nitride frameworks for artificial photosynthesis of hydrogen peroxide, *Adv. Mater.* 35 (2023) 2306831.
- [22] R. Li, M. Zheng, X. Zhou, D. Zhang, Y. Shi, C. Li, M. Yang, Carbon vacancies in porous g-C₃N₄ nanosheets induced robust H₂O₂ production for highly efficient photocatalysis-self-Fenton system for metronidazole degradation, *Chem. Eng. J.* 464 (2023) 142584.
- [23] Z. Chen, A. Savateev, S. Pronkin, V. Papaefthimiou, C. Wolff, M.G. Willinger, E. Willinger, D. Neher, M. Antonietti, D. Dontsova, “The easier the better” preparation of efficient photocatalysts—metastable poly(heptazine imide) salts, *Adv. Mater.* 29 (2017) 1700555.
- [24] Y. Yang, G. Zeng, D. Huang, C. Zhang, D. He, C. Zhou, W. Wang, W. Xiong, X. Li, B. Li, W. Dong, Y. Zhou, Molecular engineering of polymeric carbon nitride for highly efficient photocatalytic oxytetracycline degradation and H₂O₂ production, *Appl. Catal. B: Environ. Energy* 272 (2020) 118970.
- [25] I. Krivtsov, D. Mitoraj, C. Adler, M. Ilkaeva, M. Sardo, L. Mafra, C. Neumann, A. Turchanin, C. Li, B. Dietzek, R. Leiter, J. Biskupek, U. Kaiser, C. Im, B. Kirchoff, T. Jacob, R. Beranek, Water-soluble polymeric carbon nitride colloidal nanoparticles for highly selective quasi-homogeneous photocatalysis, *Angew. Chem. Int. Ed.* 59 (2020) 487–495.
- [26] B. Jürgens, E. Irran, J. Senker, P. Kroll, H. Müller, W. Schnick, Melem (2,5,8-Triamino-tri-s-triazine), an Important Intermediate during condensation of melamine rings to graphitic carbon nitride: synthesis, structure determination by x-ray powder diffractometry, solid-state NMR, and theoretical studies, *J. Am. Chem. Soc.* 125 (2003) 10288–10300.
- [27] G. Zhang, C. Huang, X. Wang, Dispersing molecular cobalt in graphitic carbon nitride frameworks for photocatalytic water oxidation, *Small* 11 (2015) 1215–1221.
- [28] H. Yu, R. Shi, Y. Zhao, T. Bian, Y. Zhao, C. Zhou, G.I.N. Waterhouse, L.-Z. Wu, C.-H. Tung, T. Zhang, Alkali-assisted synthesis of nitrogen deficient graphitic carbon nitride with tunable band structures for efficient visible-light-driven hydrogen evolution, *Adv. Mater.* 29 (2017) 1605148.
- [29] Y. Wang, P. Du, H. Pan, L. Fu, Y. Zhang, J. Chen, Y. Du, N. Tang, G. Liu, Increasing solar absorption of atomically thin 2D carbon nitride sheets for enhanced visible-light photocatalysis, *Adv. Mater.* 31 (2019) 1807540.
- [30] V.W.-h. Lau, I. Moudrakovski, T. Botari, S. Weinberger, M.B. Mesch, V. Duppel, J. Senker, V. Blum, B.V. Lotsch, Rational design of carbon nitride photocatalysts by identification of cyanamide defects as catalytically relevant sites, *Nat. Commun.* 7 (2016) 12165.
- [31] B.-X. Zhou, S.-S. Ding, K.-X. Yang, J. Zhang, G.-F. Huang, A. Pan, W. Hu, K. Li, W.-Q. Huang, Generalized synthetic strategy for amorphous transition metal oxides-based 2D heterojunctions with superb photocatalytic hydrogen and oxygen evolution, *Adv. Funct. Mater.* 31 (2021) 2009230.
- [32] C. Gan, M. Sheng, Z. Hu, Y. Li, Y. Peng, Z. Xiang, B. Sun, H. Jiang, Novel and efficient strategy for chlorophenols and CO₂ transformation over carbon nitride nanotubes: Effect of the hydroxyl grafting and surface electron polarization, *J. Catal.* 413 (2022) 636–647.
- [33] Z. Fang, Y. Bai, L. Li, D. Li, Y. Huang, R. Chen, W. Fan, W. Shi, In situ constructing intramolecular ternary homojunction of carbon nitride for efficient photoinduced molecular oxygen activation and hydrogen evolution, *Nano Energy* 75 (2020) 104865.
- [34] X. Li, W. Bi, L. Zhang, S. Tao, W. Chu, Q. Zhang, Y. Luo, C. Wu, Y. Xie, Single-Atom Pt as Co-catalyst for enhanced photocatalytic H₂ evolution, *Adv. Mater.* 28 (2016) 2427–2431.
- [35] Y. Duan, Y. Wang, L. Gan, J. Meng, Y. Feng, K. Wang, K. Zhou, C. Wang, X. Han, X. Zhou, Amorphous carbon nitride with three coordinate nitrogen (N_{3C}) vacancies for exceptional NO_x abatement in visible light, *Adv. Energy Mater.* 11 (2021) 2004001.
- [36] C. Liang, H.-Y. Niu, H. Guo, C.-G. Niu, Y.-Y. Yang, H.-Y. Liu, W.-W. Tang, H.-P. Feng, Efficient photocatalytic nitrogen fixation to ammonia over bismuth monoxide quantum dots-modified defective ultrathin graphitic carbon nitride, *Chem. Eng. J.* 406 (2021) 126868.
- [37] F. Chen, L.-L. Liu, Y.-J. Zhang, J.-H. Wu, G.-X. Huang, Q. Yang, J.-J. Chen, H.-Q. Yu, Enhanced full solar spectrum photocatalysis by nitrogen-doped graphene quantum dots decorated BiO_{2-x} nanosheets: Ultrafast charge transfer and molecular oxygen activation, *Appl. Catal. B: Environ. Energy* 277 (2020) 119218.
- [38] H. Kim, C.E. Choong, I. Han, C.M. Park, I.W. Nah, J.R. Kim, B.-H. Jeon, Y. Yoon, M. Jang, Insight into the role of charge carrier mediation zone for singlet oxygen production over rod-shape graphitic carbon nitride: Batch and continuous-flow reactor, *J. Hazard. Mater.* 424 (2022) 127652.
- [39] Z. Bicheng, B. Cheng, L. Zhang, J. Yu, Review on DFT calculation of s-triazine-based carbon nitride, *Carbon Energy* 1 (2019).

- [40] X. Xiao, Y. Gao, L. Zhang, J. Zhang, Q. Zhang, Q. Li, H. Bao, J. Zhou, S. Miao, N. Chen, J. Wang, B. Jiang, C. Tian, H. Fu, A promoted charge separation/transfer system from Cu single atoms and C₃N₄ layers for efficient photocatalysis, *Adv. Mater.* 32 (2020) 2003082.
- [41] D. Chen, K. Wang, W. Hong, R. Zong, W. Yao, Y. Zhu, Visible light photoactivity enhancement via CuTCPP hybridized g-C₃N₄ nanocomposite, *Appl. Catal. B: Environ. Energy* 166–167 (2015) 366–373.
- [42] X. Bai, L. Wang, Y. Wang, W. Yao, Y. Zhu, Enhanced oxidation ability of g-C₃N₄ photocatalyst via C₆₀ modification, *Appl. Catal. B: Environ. Energy* 152–153 (2014) 262–270.
- [43] X. Zhang, X. Xie, H. Wang, J. Zhang, B. Pan, Y. Xie, Enhanced Photoresponsive Ultrathin Graphitic-Phase C₃N₄ Nanosheets for Bioimaging, *JACS* 135 (2013) 18–21.
- [44] H. Ou, P. Yang, L. Lin, M. Anpo, X. Wang, Carbon nitride aerogels for the photoredox conversion of water, *Angew. Chem. Int. Ed.* 56 (2017) 10905–10910.
- [45] Y. Shiraishi, Y. Kofuji, H. Sakamoto, S. Tanaka, S. Ichikawa, T. Hirai, Effects of surface defects on photocatalytic H₂O₂ production by mesoporous graphitic carbon nitride under visible light irradiation, *ACS Catal.* 5 (2015) 3058–3066.
- [46] L. Yang, G. Dong, D.L. Jacobs, Y. Wang, L. Zang, C. Wang, Two-channel photocatalytic production of H₂O₂ over g-C₃N₄ nanosheets modified with perylene imides, *J. Catal.* 352 (2017) 274–281.
- [47] S.C. Perry, D. Pangotra, L. Vieira, L.-I. Csepei, V. Sieber, L. Wang, C. Ponce de León, F.C. Walsh, Electrochemical synthesis of hydrogen peroxide from water and oxygen, *Nat. Rev. Chem.* 3 (2019) 442–458.
- [48] Z. Tian, C. Han, Y. Zhao, W. Dai, X. Lian, Y. Wang, Y. Zheng, Y. Shi, X. Pan, Z. Huang, H. Li, W. Chen, Efficient photocatalytic hydrogen peroxide generation coupled with selective benzylamine oxidation over defective ZrS₃ nanobelts, *Nat. Commun.* 12 (2021) 2039.
- [49] C. Xia, J.Y. Kim, H. Wang, Recommended practice to report selectivity in electrochemical synthesis of H₂O₂, *Nat. Catal.* 3 (2020) 605–607.
- [50] K. Gong, F. Du, Z. Xia, M. Durstock, L. Dai, Nitrogen-doped carbon nanotube arrays with high electrocatalytic activity for oxygen reduction, *Science* 323 (2009) 760–764.
- [51] C. Liu, H. Li, F. Liu, J. Chen, Z. Yu, Z. Yuan, C. Wang, H. Zheng, G. Henkelman, L. Wei, Y. Chen, Intrinsic activity of metal centers in metal–nitrogen–carbon single-atom catalysts for hydrogen peroxide synthesis, *J. Am. Chem. Soc.* 142 (2020) 21861–21871.
- [52] C. He, J. Lei, X. Li, Z. Shen, L. Wang, J. Zhang, Proton-enriched alginate-graphene hydrogel microreactor for enhanced hydrogen peroxide photosynthesis, *Angew. Chem. Int. Ed. Engl.* 63 (2024) e202406143.
- [53] M. Liu, M. Gao, L. Pei, Y. Ji, X. Gu, H. Wang, H. Tan, J. Zhao, J. Jia, Z. Zheng, Tailoring phenol photomineralization pathway over polymeric carbon nitride with cyano group multifunctional active sites, *Appl. Catal. B: Environ. Energy* 284 (2021) 119710.
- [54] V.P. Nicu, J. Autschbach, E.J. Baerends, Enhancement of IR and VCD intensities due to charge transfer, *Phys. Chem. Chem. Phys.* 11 (2009) 1526–1538.
- [55] L. Zhang, Z. Jin, N. Tsubaki, Activating and optimizing the MoS₂@MoO₃ S-scheme heterojunction catalyst through interface engineering to form a sulfur-rich surface for photocatalyst hydrogen evolution, *Chem. Eng. J.* 438 (2022) 135238.
- [56] X. Zhang, B. Xu, S. Wang, X. Li, C. Wang, Y. Xu, R. Zhou, Y. Yu, H. Zheng, P. Yu, Y. Sun, Carbon nitride nanotubes anchored with high-density CuN_x sites for efficient degradation of antibiotic contaminants under photo-Fenton process: Performance and mechanism, *Appl. Catal. B: Environ. Energy* 306 (2022) 121119.
- [57] X. Li, Y. Hu, C. Zhang, C. Xiao, J. Cheng, Y. Chen, Electro-activating of peroxymonosulfate via boron and sulfur co-doped macroporous carbon nanofibers cathode for high-efficient degradation of levofloxacin, *J. Hazard. Mater.* 442 (2023) 130016.
- [58] B. Li, C. Wang, N. Li, T. Liu, X. Wang, Hydrophobic microenvironment mediated photo-Fenton beads confining free radicals in vicinity of water-soluble contaminants for enhancing water purification, *J. Clean. Prod.* 434 (2024) 140135.
- [59] B. Li, C. Wang, N. Li, C. Chen, Z. Zhu, X. Tang, Y. Cui, T. Liu, C.K. Attatsi, X. Wang, Partially oxidized mackinawite/biochar for photo-Fenton organic contaminant removal: Synergistically improve interfacial electron transfer and H₂O₂ activation, *Environ. Pollut.* 346 (2024) 123660.
- [60] Z. Zhu, J. Ye, X. Tang, Z. Chen, J. Yang, P. Huo, Y.H. Ng, J. Crittenden, Vacancy-Rich CoS_x@LDH@Co-NC Catalytic Membrane for Antibiotic Degradation with Mechanistic Insights, *Environ. Sci. Technol.* 57 (2023) 16131–16140.
- [61] C. Dai, S. Li, Y. Duan, K.H. Leong, S. Liu, Y. Zhang, L. Zhou, Y. Tu, Mechanisms and product toxicity of activated carbon/peracetic acid for degradation of sulfamethoxazole: implications for groundwater remediation, *Water Res.* 216 (2022) 118347.
- [62] B. Zhang, Y. He, W. Shi, L. Liu, L. Li, C. Liu, P.N.L. Lens, Biotransformation of sulfamethoxazole (SMX) by aerobic granular sludge: Removal performance, degradation mechanism and microbial response, *Sci. Total Environ.* 858 (2023) 159771.